

10. Answer: A

The higher off the Earth's surface we go, the less negative is the value of the gravitational potential, since gravitational potential is 0 at an infinite distance from Earth. Hence, Options C and D are out.

Gravitational potential ($\phi = -\frac{GM}{r}$) is inversely proportional to distance r from the centre of the Earth.

$$\frac{\phi'}{\phi} = \frac{r_E}{r'}$$

$$\phi' = \frac{r_E}{r'} \times \phi$$

$$= \frac{6.371 \times 10^6}{6.371 \times 10^6 + 50000} \times (-6.257 \times 10^7)$$

$$= -6.208 \times 10^7 \text{ J kg}^{-1}$$